Calcurion Maths Worksheets

GCSE Mathematics

Topic: Equations & Inequalities

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $3x + 7 = 19$. **[2]**

2. Solve $5x - 4 = 2x + 11$. **[3]**

3. Solve $\frac{x}{3} + 2 = 7$. **[2]**

4. Solve $2(x - 3) = 10$. **[2]**

5. Solve $3(2x + 1) = 2(x + 5)$. **[3]**

6. Solve $\frac{2x - 1}{3} = 5$. **[3]**

7. Solve the simultaneous equations $x + y = 7$, $x - y = 1$. **[3]**

8. Solve the simultaneous equations $2x + y = 8$, $x - y = 1$. **[3]**

9. Solve the simultaneous equations $3x + 2y = 16$, $x + y = 5$. **[4]**

10. Solve $x + 4 > 9$. **[2]**

11. Solve $2x - 3 \leq 7$. **[2]**

12. Solve $5 - 3x > -4$. **[3]**

13. Solve $x^2 - 5x + 6 = 0$. **[3]**

14. Solve $2x^2 - 5x - 3 = 0$. **[4]**

15. Represent $x \geq 2$ on a number line. **[2]**

Section B – Hard (Q16–25)

16. Solve $\dfrac{x+1}{2} + \dfrac{x-1}{3} = 4$. **[4]**

17. Solve the simultaneous equations $2x + 3y = 13$, $3x - y = 3$. **[4]**

18. Solve the simultaneous equations $4x - 3y = 11$, $3x + 2y = 4$. **[5]**

19. Solve $2x + 1 < 5x - 8$. **[3]**

20. Solve $3 \leq 2x + 1 < 9$. **[4]**

21. Solve $x^2 - 4x - 12 > 0$. **[4]**

22. Solve $x^2 + x - 6 \leq 0$. **[4]**

23. Find the integer values of x that satisfy $-3 < 2x - 1 \leq 5$. **[4]**

24. Solve $\dfrac{2}{x+1} = \dfrac{3}{x-2}$. **[4]**

25. The sum of two numbers is 15 and their difference is 3. Find the numbers. **[4]**

End of Worksheet • *Show all working clearly.*